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Microbial Infections of Respiratory system

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Objectives (summary)

□ **Upon completion of this section, you will be able to:**

- Define microbial infections of the respiratory system and explain their public health significance.
- Differentiate between upper and lower respiratory tract infections based on clinical presentation and etiology.
- Identify the major bacterial, viral, fungal, and atypical pathogens causing respiratory tract infections.
- Identify morphology, antigen structures and virulence factors of Microbial RTI
- Explain the pathogenesis, transmission, clinical manifestations, and risk factors associated with common respiratory infections.
- Discuss the laboratory diagnosis, including specimen collection, prevention and control measures, of RTIs.

Introductions to Respiratory System

- Respiratory microbial infections are the most common type of human infection caused by bacteria, viruses, fungi, or other microorganisms affecting the upper or lower respiratory tract.
- It is a major global health challenge due to their high transmissibility, substantial disease burden, pandemic potential, and increasing antimicrobial resistance.

Introductions to.....

- **The upper respiratory system** consists of the nose, pharynx and associated structures, such as the middle ear and auditory tubes-
continuously exposed to potential pathogens.
 - The normal microbiota of the nasal cavity and throat can include pathogenic microorganisms in a carrier status.
- **The lower respiratory system** consists of the larynx, trachea, bronchial tubes and alveoli, and **is usually sterile because of the action of the ciliary escalator.**
 - ✚ **Note:** Pathogens that enter the respiratory system may infect other parts of the body by hematogenous spread (Ex: septicemia, meningitis, distant focal infection).

II. Microbial Diseases of the Respiratory System

□ The infections of the respiratory system may be caused by several bacteria, fungi and viruses, and most are self-limiting.

- Pharyngitis – sore throat
- Tonsillitis – inflamed tonsils
- Sinusitis – infected sinus (nasal cavity)
- Laryngitis – infected larynx
- Epiglottitis – *H. influenzae type b* can cause epiglottitis.
- Tuberculosis
- Pertussis (whooping cough)
- Pneumonia

infections of LRT



Diseases of the Respiratory System.....

- Respiratory pathogens are circulate within a community and easily transmitted from human to human.
- Some respiratory pathogens exist as part of the normal flora.
- Others are acquired from animal source, water, air etc.
- Some pathogens are restricted to certain sites.
 - *Legionella* only infects the lung.
- Other pathogens cause infection in multiple sites.
 - *Streptococcus* can cause:
 - Middle ear infections.
 - Sinusitis.
 - Pneumonia, etc.

III. Important Etiologic Agents of Respiratory Infections

A. Bacterial Pathogens

- *Streptococcus pneumoniae*
- *Streptococcus pyogenes*
- *Haemophilus influenzae*
- *Staphylococcus aureus*
- *Klebsiella pneumoniae*
- *Mycobacterium tuberculosis*
- *Pseudomonas aeruginosa*
- *Bordetella pertussis*
- Inhalation anthrax

B. Atypical Bacteria

- *Mycoplasma pneumoniae*
- *Chlamydia pneumoniae*
- *Legionella pneumophila*

C. Viral Pathogens

- Influenza viruses
- Respiratory syncytial virus (RSV)
- Rhinoviruses
- Adenoviruses
- Coronaviruses
- Human metapneumovirus
- Parainfluenza viruses

D. Fungal Pathogens

- *Aspergillus* spp.
- *Pneumocystis jirovecii*
- *Histoplasma capsulatum*
- *Cryptococcus neoformans*

1. Streptococci *Species*

General Characteristics

- ➡ Are Gram-positive, non-motile, non-spore forming, **Catalase-negative cocci** that occur in pairs or chains
- ➡ Most are facultative anaerobes, and some are obligate (strict) anaerobes
- ➡ Most require enriched media (blood agar)
- ➡ Most are normal flora of **upper respiratory tract, gastrointestinal tract and genitourinary tract**

Classification of Streptococci

- Heterogeneous group of bacteria; No system suffice to classify
 - The current classification depends on:
 - Colony morphology & Hemolytic reaction
 - Serological (Lancefield) grouping
 - Biochemical reaction

1. Based on hemolytic pattern on blood agar;

- **Alpha (α) -Hemolysis:** partial lysis of RBCs, e.g. *S. pneumoniae* & *S. viridans*
- **Beta (β) -Hemolysis:** complete lysis of RBCs, e.g. Group A & B (*S. pyogenes* & *S. agalactiae*)
- **Gamma (γ) -hemolysis:** no lysis of RBCs, e.g. Group D (*Enterococcus spp*)

Classification.....

2. Serologic (Lancefield) groupings: this is based on cell wall antigens there are 21 Lancefield groupings; A – H, K – M, O – V

- Groups A, B and D are common human pathogens

e.g., *S. pyogenes* grouped under Group A Streptococcus and *S. agalactiae* comes under Group B Streptococcus

3. Biochemical Properties: used for species **not be classified into the Lancefield classification**

- Growth in 6.5% NaCl so/n (high salt/adverse conditions)
- Sensitivity to Bacitracin or Optochin
- Bile solubility: some *Strep.* bacteria can grow in bile

Classification ...

Species	Lancefield group	Heamolysis	Dx feature
<i>S. pyogenes</i>	A	β	Bacitracin - sensitive
<i>S. agalactiae</i>	B	β	Bacitracin- resistance
<i>S. faecalis</i>	D	Gamma	Growth in 6.5% NaCl
<i>S. bovis</i>	D	Gamma	No growth in 6.5% NaCl
<i>S. pneumonia</i>	Not applicable	Alpha	Bile soluble, inhibited by optochin
Viridians group	Not applicable	Alpha	Not Bile soluble, not inhibited by optochin

Pathogenesis of *Streptococci*

- Most pathogenic streptococci are β hemolytic and are generally referred to as “**hemolytic streptococci**”
- Cause disease when they gain access to tissue or blood
- *S. pyogenes* is found on the **skin** and in the **oropharynx** in small number
- *S. agalactiae* found in the female genital tract

Pathogenesis of Streptococci ...

- Most of the 'viridians' group (e.g., *S. mutans*, *S. sanguis*, *S. mitis*) and *S. pneumoniae* (α hemolytic streptococci) are bacterial flora found chiefly in upper respiratory tract (URT) (**oropharynx**), and are capable of causing opportunistic infections.
- The gamma streptococci comprise the group generally referred to as the '**Enterococcus**'

Group A β -Hemolytic Streptococci (*S. pyogenes*)

- *S. pyogenes* occur as long chains when recovered from liquid culture, but may appear as individual cocci, pairs, or clusters
- Nasopharyngeal carriage is common especially in colder months and particularly among children.
- Its habitat is infected patients, and also normal human carriers in whom the organism resides on skin and mucous membranes.

Streptococcus pyogenes.....

- The most pathogenic of the group and the cause of **three type of disease**:
 - **Pyogenic**- pharyngitis (**sore throat**) and cellulitis
 - **Toxigenic**- scarlet fever and toxic shock syndrome
 - **Immunologic disease (Post-streptococcal diseases)** -
rheumatic fever and acute glomerulonephritis
- Produces a large number of powerful enzymes and toxins

Antigenic structure and other virulence factors

1. Components of cell wall: such as capsule, M Protein & lipoteichoic Acid-Anti-phagocytic and adherence factor

2. Toxins production:

- ❑ **Pyogenic exotoxin (erythrogenic toxin A-C):** Lyse RBCs and cause scarlet fever & toxic shock syndrome
- ❑ **Hemolysin/Streptolysin (O & S):** destroy RBCs
 - **Streptolysin O:** oxygen labile & lyse RBC in anaerobic condition, has antigenic in nature (ASO-test)
 - **Streptolysin S:** oxygen stable, responsible for hemolytic zones (on blood agar plate) and it is not antigenic

Virulence Factors.....

3. The enzymes contribute for the invasiveness includes:

- **Streptokinase (Fibrinolysin):** dissolves fibrin
- **Hyaluronidase:** degrades hyaluronic acid and it is spreading factor
- **DNase:** depolymerizes DNA in exudates or necrotic tissue

Clinical significance

- *S. pyogens* responsible for:
 - URT infections:** sore throat/*tonsillitis/pharyngitis*
 - Scarlet fever:** Strep throat characterized with a diffuse erythematous **bright red rash** due to Erythrogenic toxin
 - Ear infection:** *Acute otitis media*
 - Sepsis:** *Streptococcal toxic shock syndrome*
 - Skin infection:** soft tissue infections (impetigo, cellulitis Erysipelas,)
 - Necrotizing fascitis** (Streptococcal gangrene)
 - Post-streptococcal diseases:** (*rheumatic fever, acute glomerulonephritis*)

- **Post streptococcal (Non- Suppurative) disease:** - Is due to the inflammation caused by immunologic response to M proteins that cross – react with human tissue
 - **Rheumatic fever** –is due to cross-reaction between antibody & antigens of the human heart & joint tissue; occurs after 2 weeks of pharyngitis; It is characterized by fever, rash, carditis and arthritis
 - **Acute glomerulonephritis** – caused by immune complexes bound to glomeruli membrane; occurs 1 week after impetigo or pharyngitis—acute inflammation of renal glomeruli with edema, hypertension, hematuria, proteinuria, progressive and irreversible damage to kidneys

Epiglottitis & **Laryngitis**

□ **Epiglottitis**- Inflammation of the cartilage that covers the trachea (windpipe)-*H. influenzae*, *S. pneumoniae* or *S. pyogenes*.

- Often characterized by the abrupt onset of throat and neck pain, fever and inspiratory stridor (difficulty in moving adequate amounts of air through the larynx)
- If this disease is not treated promptly, death may result from acute airway obstruction.

Epiglottitis & Laryngitis

- **Laryngitis** is swelling and irritation (inflammation) of the voice box (larynx) that is usually associated with hoarseness or loss of voice-*Haemophilus influenzae* & *Streptococcus pneumoniae*, could be fungal and viral.
- The illness is characterized by **variable fever; inspiratory stridor; hoarse phonation**; and a **harsh, barking cough**.

Naso-Pharyngitis/Streptococcal Pharyngitis/tonsillitis

- Infection of the nose & throat characterized by irritation of nose & throat, fever, sneezing, vomiting & muscular aches
- A variety of bacteria can cause infection in the pharynx
 - Caused by *Streptococcus pyogenes*
 - Contains M proteins which inhibits phagocytosis
 - Produces pyogenic toxins which cause the symptoms seen with pharyngitis
 - Complications of naso-pharyngitis:
 - Otitis media
 - Lower respiratory tract infection
 - Older child may develop sinusitis
- Group A streptococci can cause abscesses on the tonsils.
- *S. pyogenes* can cause scarlet fever and toxic shock syndrome.

Streptococcal Pharyngitis/tonsillitis.....

- Bacteria colonize mucous membranes of pharynx and tonsils
- **Sx's:** -reddened adenoids -side of the throat (URT bacterial diseases), edema, enlargement and tenderness of throat (**severe sore throat**); painful swallowing; purulent exudates over



Scarlet fever

- Caused by Group A streptococci

- Usually seen in children under age of 18 years

- **Scarlet fever** is characterized with the appearance a diffuse erythematous (redder) rash initially on the upper chest and then spreads to the extremities and over the entire body

-armpits and groin

- Erythrogenic toxin responsible for bright red rash typical of scarlet fever

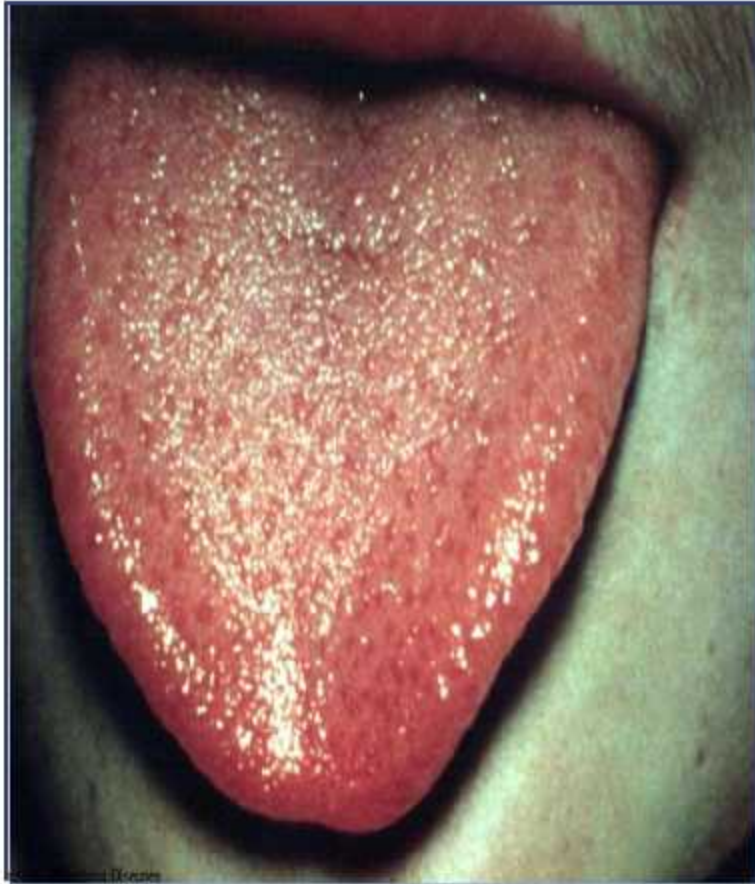
- Rash lasts 2-5 days

..Scarlet fever

- **Symptoms** can also include:
 - Very sore throat with yellow or white papules
 - Fever of 101°F or higher
 - Lymphadenopathy in neck
 - Headache, body aches, and nausea



Fig. 16.3 The characteristic erythematous rash on (A) tongue and (B) skin associated with scarlet fever.



Scarlet Fever - In the early stages there is a dense white coating. Later this peels off leaving a raw red surface, with prominent follicles, the “**strawberry tongue**”.

Clinical significance ...



Scarlet Fever Rash



Necrotizing fasciitis



Erysipelas

Otitis Media:

- It is uncomfortable infection of the middle ear.
- Earache, or otitis media, can occur as a complication of nose and throat infections.
- Pus accumulation causes pressure on the eardrum.
- Bacterial causes include *S. pneumoniae*, *Hemophilus influenzae*, *Moraxella catarrhalis*, *S. pyogenes*, and *S. aureus*.

- Sepsis (streptococcal toxic shock syndrome or toxic shock-like syndrome, TSLSS): the organism is introduced into the subcutaneous tissue through a break in the skin → cellulitis → necrotizing fasciitis → systemic toxicity, multiple organ failure, and death (mortality > 40%)

Viridians streptococci

- Produces α -hemolysis on blood agar
- Includes *S. salivarius*, *S. mutans*, *S. sanguis*, *S. mitis* and *S. milleri*.
- They are part of resident normal flora of the throat & mouth
- Though normally non pathogenic, formation of dental caries and other oral cavity (gingival / gum) infections
- They are also capable of causing **infective endocarditis**
- They usually enter the blood stream from the oropharynx, typically after dental surgery

Medically important streptococci included:-

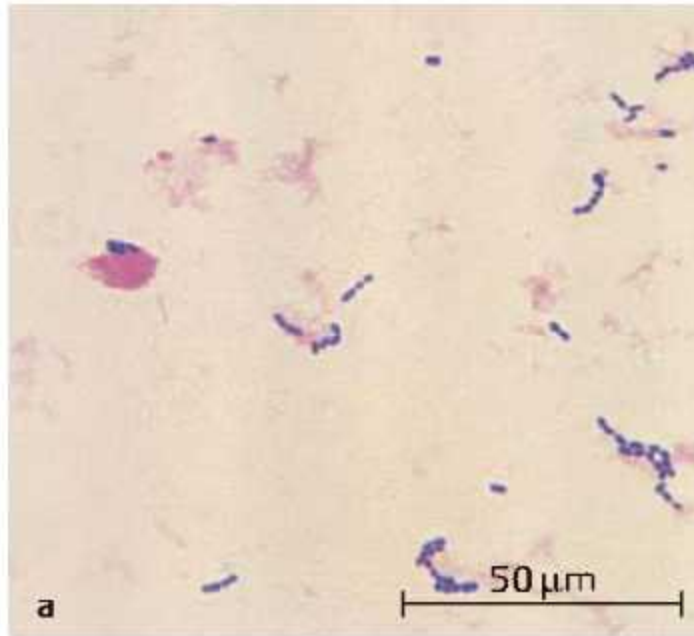
<i>S. pyogenes</i> (group A)	Pharyngitis, scarlet fever, pyoderma, erysipelas, cellulitis, necrotizing fasciitis, streptococcal toxic shock syndrome, bacteremia, rheumatic fever, glomerulonephritis
<i>S. agalactiae</i> (group B)	Neonatal infections (meningitis, pneumonia, bacteremia), urinary tract infections, amnionitis, endometritis, wound infections, pneumonia, bacteremia
Other β -hemolytic streptococci	Pharyngitis, abscess formation, bacteremia
Viridans group streptococci	Bacteremia, endocarditis, abscess formation, dental caries
<i>S. pneumoniae</i>	Pneumonia, meningitis, sinusitis, otitis media, bacteremia

Laboratory diagnosis of *Streptococci*

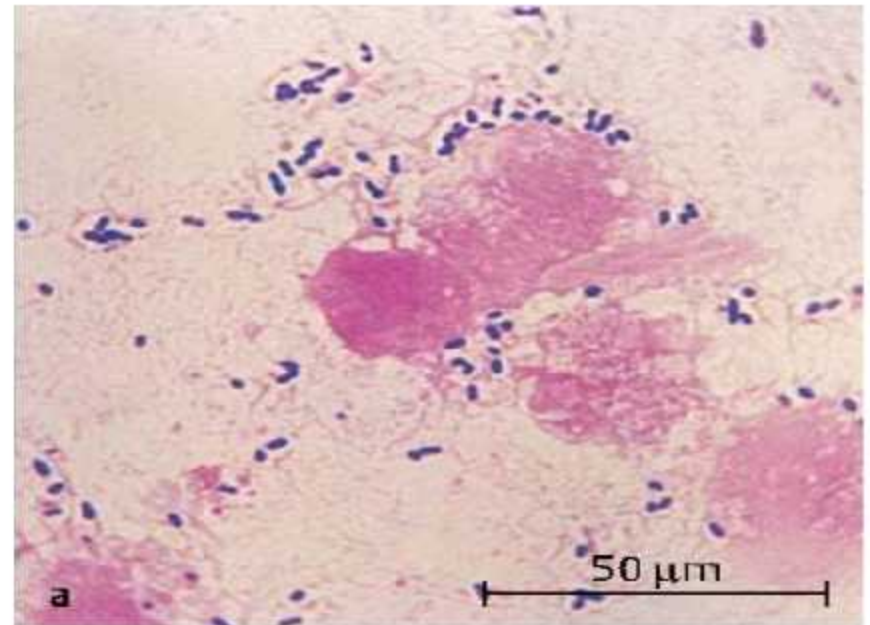
- **Specimens:** throat swab, pus, blood, ear swab, wound swab, CSF (from neonates), urine or any of the specimen depending on the site of infection
- **Grams stained smears:** All *Streptococci* spp reported as gram positive cocci in short chains, in pairs (diplococci) or singly

Laboratory diagnosis.....

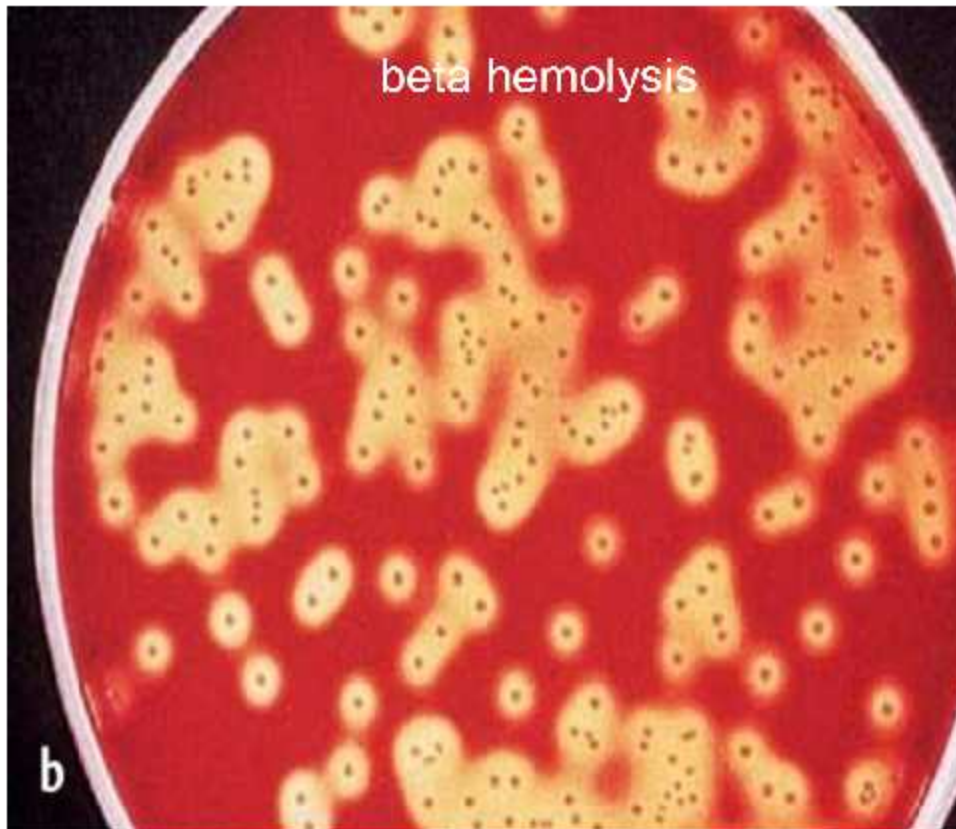
- **Culture:** majority of *streptococci* grow aerobically and anaerobically b/n 22-42°C (37°C optimum temp)
 - **Blood agar** (use 5% sheep, horse or rabbit blood)
 - *S. pyogenes* appear as **small**, colourless, dry, shiny or mucoid colonies surrounded by a large zone of beta hemolysis on sheep blood agar
 - Does not grow in MacConkey agar plate.
 - Bacitracin sensitive.
 - Pyrrolidonyl arylamidase (PYR) enzyme test-Positive
- **Rapid latex antigen kits** for direct detection of patient's antibody titer to streptolysin-O (ASO test) after group A streptococcal infection



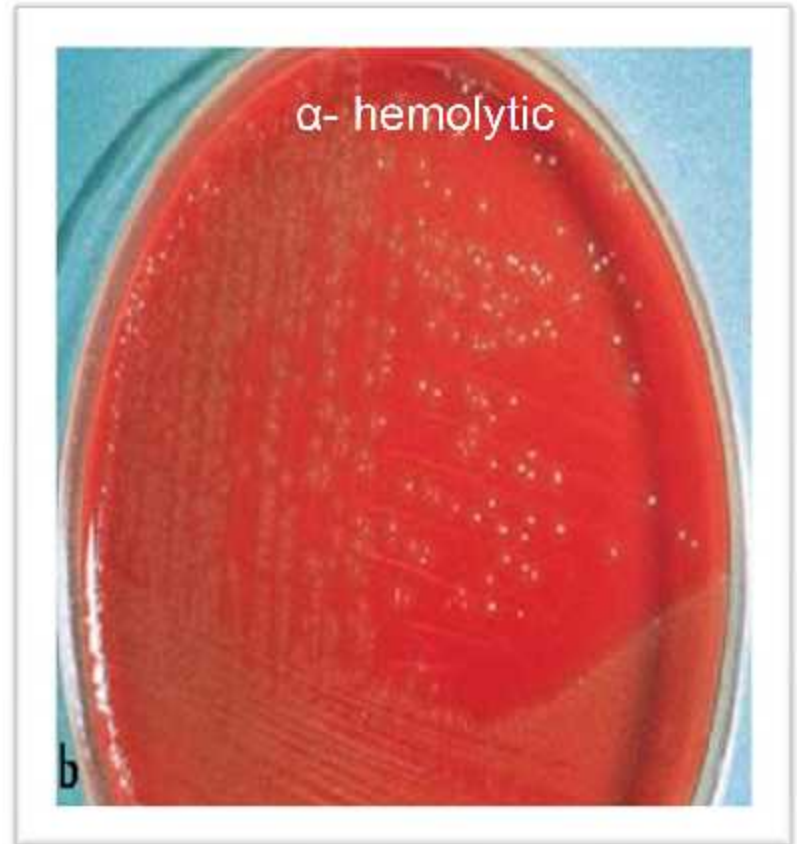
Gram staining of pleural puncture biopsy material: gram-positive cocci in twisted chains- *S. pyogenes*



Gram staining of a preparation of middle ear secretion: *Streptococcus pneumoniae*



Culture on blood agar: *S. pyogenes*

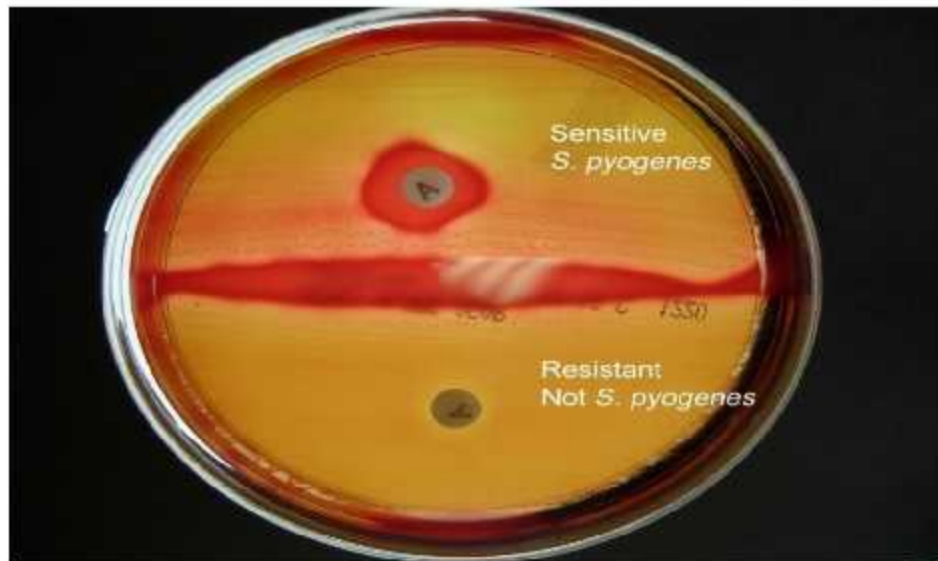


Culture on blood agar: *S. pneumoniae*

Bacitracin sensitivity test

Group A → Sensitive

Group B-U → Resistant



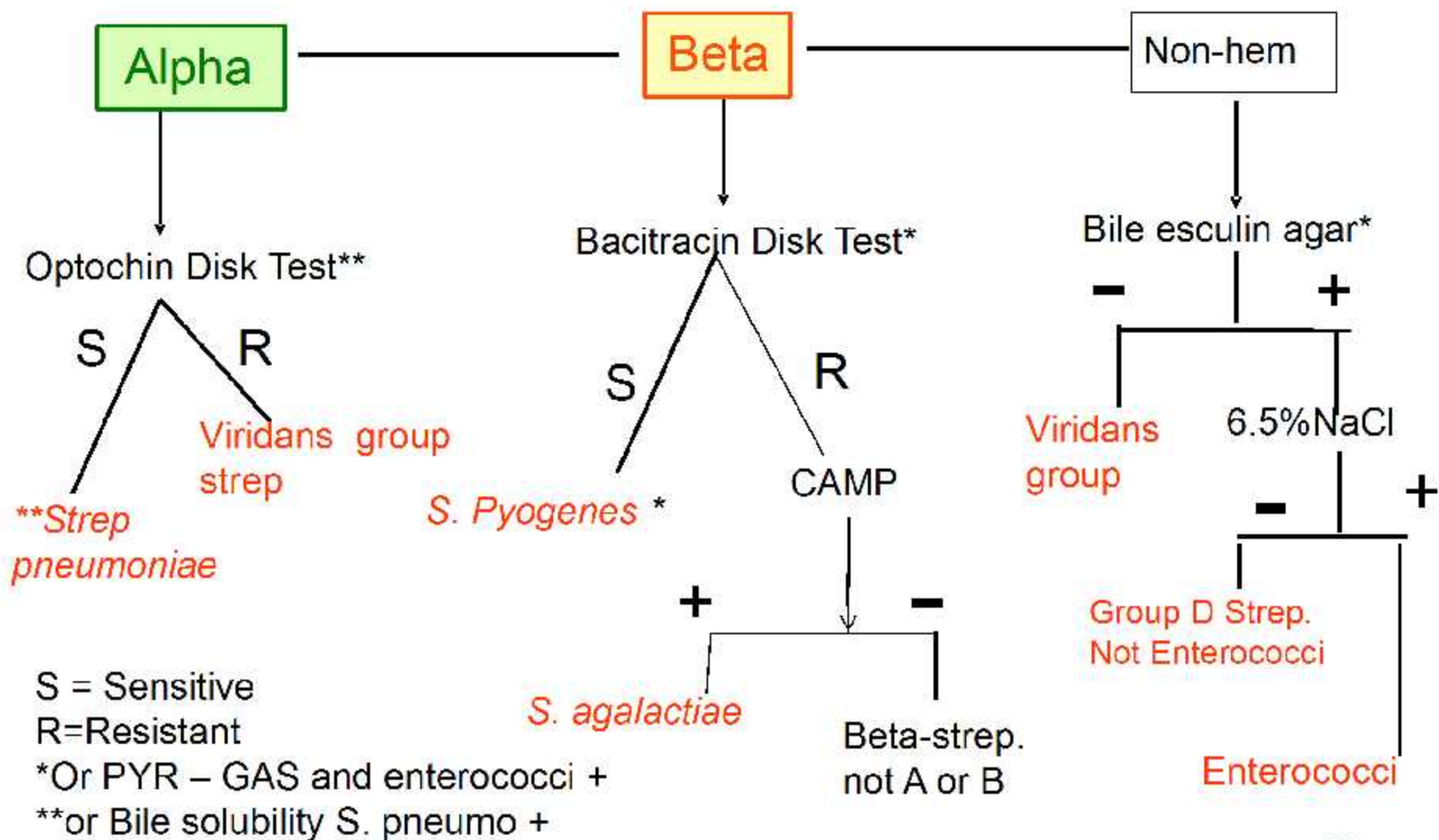
Bacitracin disk test

Laboratory diagnosis.....

- *S. pneumoniae* grows best in chocolate agar enriched with 5-10% CO₂
 - Small □ hemolytic colonies
 - Growth inhibited by optochin and the cells are lysed by bile acids (bile solubility positive)
- *Viridans streptococci*- Show greenish discoloration of blood agar plate.
 - Optochin resistant.
 - Not soluble in bile salts.
 - Do not ferment isculin.



Differentiation of Streptococci (1)



Prevention and control

- ➡ Treatment is usually with penicillin / Erythromycin
- ➡ However, strains resistant to penicillin and multiple antibiotics are rapidly emerging
- ➡ To prevent neonatal disease, it is recommended that all pregnant women should be **screened for colonization** with group B streptococci at 35 to 37 weeks of gestation
- ➡ A vaccine is available for *S. pneumoniae*

2. Diphtheria ... *Corynebacterium diphtheriae*

- Caused by the toxin produced by *Corynebacterium diphtheriae*
 - Non-motile, non-spore-forming, non-capsulated and catalase positive, aerobic and facultative anaerobe gram-positive rods
 - **Contain mycolic acid in the cell wall, however non acid fast**
 - **Irregular or pleomorphic shape (form V, Y and club shaped Chinas' letter Shape)**



Clinical significance

- **Diphtheria** is most commonly an infection of the upper respiratory tract and causes pharyngitis, fever, swelling of the neck (**bull neck diphtheria**)
- Spread by droplets, secretions, or direct contact
- Symptoms of pharyngeal diphtheria vary from localized infection of mild **pharyngitis** (sore throat) to **hypoxia** due to airway obstruction by the **plaque-like pseudomembrane in the throat**

Clinical significance.....

- Life-threatening systemic complications, principally loss of motor function (e.g., difficulty in swallowing) and congestive heart failure may develop due to the action of diphtheria toxin.
- The inflammatory reaction leads to collection of a greyish-white exudates, the matrix of the “diphtheria pseudomembrane” consisting of fibrin, dead granulocytes, and necrotic epithelial cells.
 - This coating adheres quite strongly to the mucosa.
 - It may extend into the larynx, thus eventually hindering respiration.
 - Regional lymph nodes are highly swollen



Diphtheria Membrane



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Corynebacterium diphtheriae infection of the throat. Gross swelling and congestion of the whole pharyngeal and tonsillar area, with a gray exudate covering the tonsil.

Laboratory diagnosis

- Initial treatment of a patient with diphtheria is on clinical grounds
- **Specimens:** throat, nasopharyngeal & skin swab
- **Microscopy:** Beaded rods in typical arrangement when stained with **alkaline methylene blue or gram's stain**
 - In **Albert stained smears:** *C. diphtheriae* often appears beaded due to the presence of dark staining granules in the rods
 - In **toluidine blue stained smears**, the organisms stain **pale** blue and the granules dark red-purple.
- PCR, ELISA & an immunochromographic strip assay have been described for detection of the diphtheria toxin or gene

Culture:

- *C. diphtheriae* growth is 20–40 °C with an optimum of 35–37 °C at an aerobe and facultative anaerobe.
- **On blood agar:** Small, granular, and gray, with irregular edges with small zone of hemolysis
- **Selective medium:** Loeffler agar medium, Tellurite agar medium, or Tinsdale tellurite agar are selective medium

Biochemical tests

- All *C. diphtheriae* reduces nitrate to nitrites
- *Toxigenic strain of C. diphtheriae* is:
 - Catalase, oxidase and urease negative.
 - Acid production from a range of carbohydrate fermentation, such as glucose and maltose
 - *C. diphtheriae* also ferments starch with acid production.

Biochemical reaction...

▣ Elek toxigenicity testing

- A filter paper disk containing antitoxin (10 IU/disk) is placed on an agar plate.
- The cultures to be tested for toxigenicity are inoculated 7–9 mm away from the disk.
- After 48 hours of incubation, the antitoxin diffusing from the paper disk has precipitated the toxin and has resulted in precipitin bands formations

Prevention and control

- Immunization with Toxoid (DPT)

Treatment

- Antibiotics (e.g., penicillin and erythromycin)
- Active immunization in childhood with diphtheria toxoid yields antitoxin levels that are generally adequate until adulthood
- Young adults should be given boosters of toxoid

VI. Bacterial Diseases of the Lower Respiratory System

- Many of the same microorganisms that infect the upper respiratory system also infect the lower respiratory system.
- Diseases of the lower respiratory system include:
 - Bronchitis
 - Pneumonia
 - Tuberculosis, and
 - Pertussis (whooping cough)

Bacterial Diseases of the LRT.....

- **Acute bronchitis** are usually self-limited viral infections in bronchus that should not be treated with antimicrobials.
 - Characterized by inflammation of the bronchi.
 - Clinically expressed as cough, usually with sputum, and evidence of concurrent upper respiratory infection.
- **Pneumonias** are usually bacterial infections in lung that require antimicrobial treatment
 - *Pneumonia* is a term applied to severe complications of bronchitis with the alveoli involved.

Bacterial Diseases of the LRT.....

- Bacterial pneumonia
- *Chlamydial pneumonia*
- *Mycoplasma pneumonia*
- *Tuberculosis*
- Pertussis
- Inhalation anthrax
- *Legionella pneumonia* (Legionnaire's disease)
- Q fever
- Psittacosis (Ornithosis)

1. *Streptococcus pneumoniae*

- Encapsulated, non-motile, lancet shaped gram- positive cocci in pairs (diplococci) or short chains
- It is aerobic & facultative anaerobe, result in □-hemolysis on blood agar
- Do not grow on MacConkey agar and catalase negative



***Streptococcus pneumoniae*.....**

- More than 85 antigenic types of *S. pneumoniae*, based on capsule polysaccharide antigens
- ***Pneumococci* responsible for;**
 - Community acquired pneumonia
 - Sinusitis and otitis media
 - Meningitis, and bacteremia

Pathogenesis and clinical pictures

- The virulence of *S. pneumoniae* is its ability to produce
 - Leucocidin
 - Hyaluronidase, Pneumolysins
 - Capsule which prevents phagocytosis
- The capsule protects the pathogens from phagocytosis and is the most important determinant of pneumococcal virulence.
- Unencapsulated variants are not capable of causing disease.
- Other potential virulence factors include pneumolysin with its effects on membranes and an IgA1 protease.

Pathogenesis and clinical pictures.....

- The natural habitat of pneumococci is provided by the mucosa of the upper respiratory tract.
- Most infections are caused by **endogenous spread** from the colonized **nasopharynx or oropharynx** to distal site (e.g., lungs, sinuses, ears, blood, meninges)
- About **40–70%** of healthy adults are carriers.
- Person-to-person spread through infectious droplets is rare

Pneumococci...

- Predisposing factors include primary cardiopulmonary diseases, previous infections (e.g., influenza), and extirpation (removal) of the spleen or complement system defects.
- The most important pneumococcal infections are lobar pneumonia and bronchopneumonia.
- Other infections include acute exacerbation of chronic bronchitis, otitis media, sinusitis, meningitis, and corneal ulcer.
- Severe pneumococcal infections frequently involve sepsis.

Pneumococci...

- “Community-acquired pneumonia” – Bacterial (80–90%)

- *Streptococcus pneumoniae* (30%)
- *Haemophilus influenzae* (5%)
- *Staphylococcus aureus* (5%)
- *Klebsiella pneumoniae*
- *Legionella pneumophila*
- Mixed anaerobic flora (aspiration pneumonia)
- *Mycoplasma pneumoniae* (10%)
- *Coxiella burnetii*
- *Chlamydia psittaci*
- *Chlamydia pneumoniae*

Pathogens Most Commonly Involved with Community-Acquired Pneumonia

Bacterial

- Gram-negative bacilli 50-70%
 - ❖ *Pseudomonas aeruginosa**
 - ❖ *Enterobacteriaceae**
- *Staphylococcus aureus** 15-30%
- Anaerobic bacteria 10-30%
- *Haemophilus influenzae* 10-20%
- *Streptococcus pneumoniae* 10-20%
- Legionella* 4%

Viral

- Cytomegalovirus
- Influenza virus*
- Respiratory syncytial virus*

Fungal

- Aspergillus*

10-20%

<1%

*May cause nosocomial epidemics

Pneumococci...

- **“Hospital-acquired pneumonia” mostly caused by:**
 - Enterobacteriaceae
 - Pseudomonas aeruginosa
 - Staphylococcus aureus
- Empyema
 - Streptococcus pneumoniae
 - Staphylococcus aureus
 - Streptococcus pyogenes
 - Numerous other bacteria are potential pathogens

LRTI... cont'd.....Acute bronchitis

- Important etiologies of **Acute bronchiolitis (small children)**
 - *Mycoplasma pneumoniae*
 - *Chlamydia pneumoniae*
- **Pertussis** *is caused by Bordetella pertussis*
- Acute exacerbation of “chronic obstructive pulmonary disease” (COPD) caused by:
 - *Streptococcus pneumoniae*
 - *Haemophilus influenzae*
 - *Moraxella catarrhalis*

RISK FACTORS FOR ACUTE RESPIRATORY INFECTIONS

- Smoking – active and passive
- Air pollution – indoor and outdoor
- Crowding e.g family size, day care centers, refugee camps, poor housing
- Malnutrition
- Low birth weight
- Acute lower respiratory infections @ < 2 years
- Socioeconomic status

Risk Factors for Community-Acquired Pneumonia (1 of 2)

1. Old age
2. Smoking (>20 cigarettes/day)
3. Air pollution
4. Chronic diseases (e.g., diabetes, chronic hepatopathies, renal, cardiac, respiratory failure)
5. Malnutrition, precarious social and economic situations
6. Acute and chronic alcoholism
7. Chronic obstructive pulmonary disease
8. Primitive ciliary dyskinesia, bronchiectasis, cystic fibrosis
9. Congestive cardiomyopathies
10. Neuromuscular diseases
11. Dementia
12. Autoimmune diseases (LES, rheumatoid arthritis, and other collagen diseases)
13. Malignancy
14. Immunodeficiencies
15. Splenectomy
16. Immunosuppressive therapies
17. Drug addiction
18. Inadequate use of antibiotics

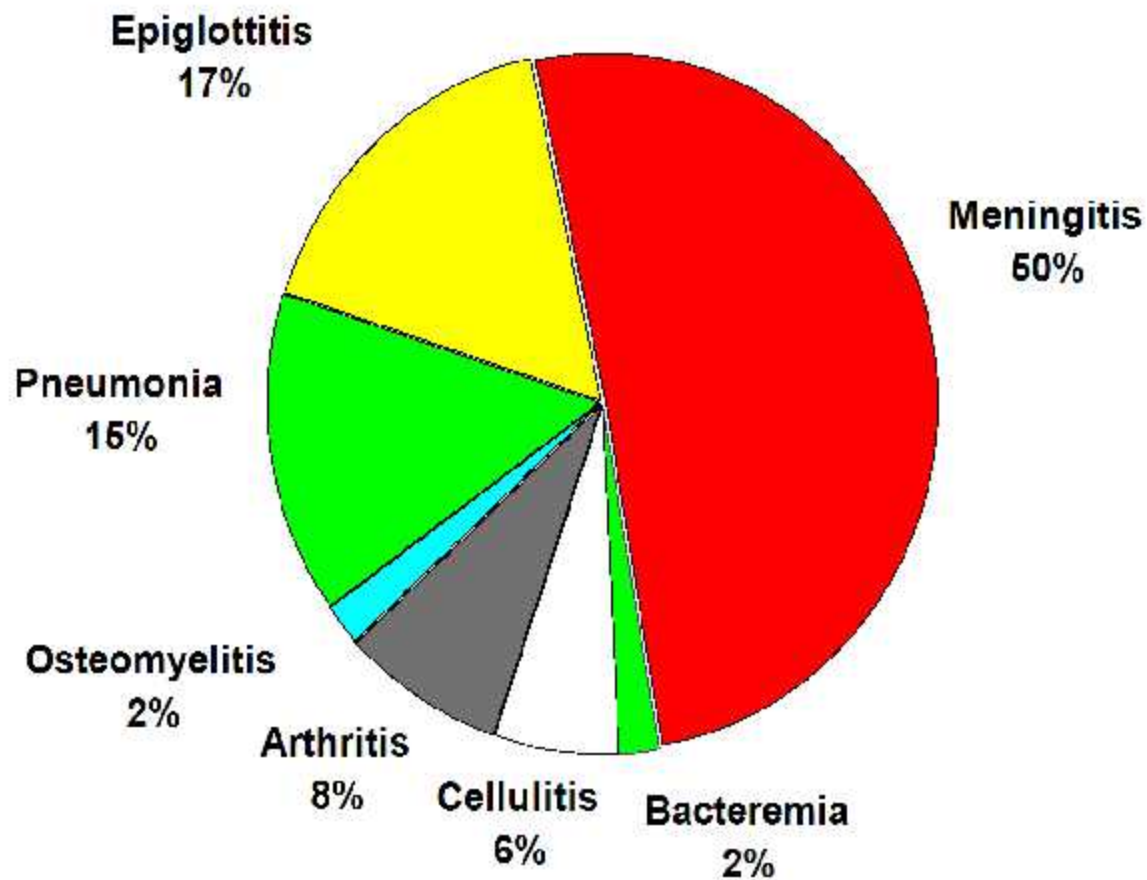
2. Haemophilus influenzae

- *H. influenzae* is a normal flora of the upper respiratory tract
- Humans are the only natural hosts, and colonization begins shortly after birth and infect only human
- Leading cause of **bacterial meningitis** in infants and very young children (6 months to 1yr)
- Otitis media, sinusitis/cellulitis, epiglottitis, arthritis osteomyelitis, bacteremia, pneumonia, conjunctivitis

Haemophilus influenzae.....

- Obligate parasites, and is fastidious Gram-negative coccobacilli
- Six serotype based on capsule antigen (a-f)
- 95% of infections caused by *H. influenzae* type b
- *H. influenzae* type b contain polyribitol phosphate (PRP) capsule that responsible for serious systemic infection
- *H. influenzae* type b requires hemin (factor X) and NAD⁺ (factor V) for growth
- Other species require only NAD⁺ (factor V)

Clinical significance



Laboratory diagnosis

- **Sample:** CSF, synovial fluid, LRT specimen, direct needle aspiration for sinusitis, otitis, nasopharyngeal swabs, pus, blood, and spinal fluid for smears and culture
- Specimens for detection of *H. ducreyi* should be collected with a moistened swab from the base or margin of the ulcer
- **Microscopy:** gram's – ve coccobacili (stained with 1% carbon fuchsin)
- Specimens must be cultured as soon as possible and not refrigerated.

Culture:

- Media must contain iron-containing haemin or porphyrin (as factor X) and Nicotinamide Adenine Dinucleotide (NAD) or its phosphate (NADP) (as factor V).
- Growth is best achieved in a moist carbon dioxide enriched atmosphere at a temperature of 20–40 °C with an optimum of 35–37 °C.

Culture:.....

□ On chocolate agar

□ *H. influenzae* grows well on chocolate agar because it contains factors X and V and produce mucoid colonies with distinctive smell.

□ Heating blood agar to 75 °C inactivates serum NADase and releases extra factor V from the red cells.

- Addition of bacitracin (300 mg/litre) provides a selective medium to recover *H. influenzae* from sputum.
- This is not needed when culturing CSF.

Biochemical test

- **Biochemical test** is not routinely done; **Satellitism test** is used to identify *H. influenzae*

Satellitism test

- This is a routine test in identification of *H. influenzae* using *S. aureus* that streaked across a plate of Blood Agar and Nutrient Agar
- After overnight incubation, *H. influenzae*: shows growth on the blood agar plate along side of the streak of *S. aureus* but not on the nutrient agar plate,
- If satellite colonies are present on both plates the organism is probably an *Haemophilus* species that requires only factor V, such as *H. parainfluenzae*.

Prevention and control

- Active immunization with conjugated **polyribitol phosphate (PRP)** given to children younger than age 2 years
- Currently **pentavalent vaccine** is available
- Rifampin is given prophylactically to individuals in close contact with a patients infected with *H. influenzae*
- **Treatment** : by ceftriaxone

3. Pertussis (whooping cough)

- Caused by *Bordetella pertussis* w/c is extremely small, strict aerobic, non- motile, Capsulated, gram–ve coccobacilli
 - Pertussis (whooping cough) in humans w/c is a highly contagious disease that occurs primarily in infants and young children
- Transmission is by droplets and colonize only ciliated cells of respiratory mucosa and they multiply rapidly

Pertussis (whooping cough).....

- After an incubation period of about 2 weeks, the "**catarrhal stage**" develops, with mild coughing and sneezing.
 - At this stage, large numbers of organisms are sprayed in droplets, and the patient is highly infectious but not very ill.
- **Paroxysmal" stage**, the cough develops its explosive character "whoop" upon inhalation.
 - This leads to rapid exhaustion and may be associated with vomiting, cyanosis, and convulsions.
- After 3-4 weeks the disease enters **Convalescent stage**
 - Frequency and severity of the coughing gradually decrease
 - But secondary complications can occur

Laboratory diagnosis

- **Specimen:** saline nasal wash and nasopharyngeal swab
- **Smear:** small, capsulated, gram-ve coccobacilli
 - With **toluidine blue stain**, bipolar metachromatic granules can be seen
- **Culture** – On selective and enrichment medium such as **chocolate-cephalexin blood agar**, gray, shiny, mercury like mucoid colonies at 37 °C for 7 days
- **Biochemical tests:** It produces a pigment in the medium and is able to grow aerobically on blood agar and nutrient agar.
 - Oxidase and Catalase positive

Prevention and control

- DPT vaccine, erythromycin as prophylaxis

- **Treatment:** erythromycin, azithromycin

4. *Mycoplasma Pneumoniae*

- *Mycoplasmal pneumonia* is common in children and young adults; as many as 20% of cases.
- Described as **“atypical “ or “walking pneumonia”**.
- Low grade fever, non-productive cough of long 2-3 weeks
- Organism has **no cell wall** so is difficult to stain.
- *M. pneumoniae* produces small colonies after 2 weeks' incubation on enriched media containing horse serum and yeast extract.
- **A complement-fixation test**, used to diagnose the disease, is based on the rising of antibody titer.



Mycoplasma

***Mycoplasma* species**



Scanning electron micrograph of *Mycoplasma pneumoniae*



Mycoplasma pneumoniae colonies

- Not seen with Gram stain because it lacks peptidoglycan cell walls
- Plastic, pleomorphic shape (neither rods nor cocci)
- Cell membrane is a sterol-containing lipid bilayer
- Colonies may have "fried egg" appearance
- Rarely cultured for diagnostic purposes

Mycoplasma pneumoniae

- Primary atypical pneumonia

PENICILLINS
CEPHALOSPORINS

Tetracyclines



Doxycycline

AMINOGLYCOSIDES

Macrolides



Azithromycin or

Clarithromycin

FLUOROQUINOLONES
OTHER



Radiograph of patient with *Mycoplasma pneumoniae* showing patchy consolidation in both lungs

TREATMENT

- Erythromycin or tetracycline are the usual agents used for treatment of *M. pneumoniae* infections.
- They shorten the course of infection, although eradication from the nasopharynx may take much longer.
- Azithromycin and clarithromycin are comparable to erythromycin
- Most quinolones are also active.

5. Chlamydial pneumonia

- Caused by *Chlamydia pneumoniae*:

- Causes atypical pneumonia, usually a mild respiratory infection with sore throat, cough, fever, and symptoms associated with asthma.

- Responsible for 10% of pneumonia cases

- *Chlamydia pneumoniae* is small, gram negative, obligate intracellular bacteria

Chlamydial pneumonia.....

- Cell wall contain Lipopolysaccharide (LPS) but doesn't have a peptidoglycan layer
- Non- motile and stain poorly with gram stain but readily with Giemsa stain
- They lack the ability to produce their own ATP therefore, they use host's ATP
- Multiply in the cytoplasm of the host cell and forming micro-colonies or inclusion bodies.

Epidemiology

- ❑ It is transmitted by inhaling infected aerosols and person-to-person contact.
- ❑ Infect a wide spectrum of vertebrate hosts including birds, mammals and humans
- ❑ Can cause both community-acquired and nosocomial infections
- ❑ Similar to *Mycoplasma pneumonia*.
- ❑ Tetracycline or erythromycin

6. Inhalation anthrax:

- **Pulmonary anthrax (wool sorter's disease)** is caused by *Bacillus anthracis*.
 - It is Spore-forming, Gram-positive rod
 - Form endospores – which is oval and centrally located
 - Some are non- motile and has **capsule** that is anti-phagocytic
 - Most members are saprophytic and widely distributed in natural environment (mainly in soil and water)

***Bacillus anthracis*.....**

- *B. anthracis* is the principal pathogen which causes **anthrax** w/c is primarily a disease of herbivores—goats, sheep, cattle, horses, etc
- In humans, the infection acquired by entry of spores through;
 - By inhalation of spores into the lung (**Pulmonary anthrax**)
 - Injured skin (**cutaneous anthrax**)
 - Rarely the mucous membranes (**gastrointestinal anthrax**)

Pathogenesis

- The pathogenicity of *B. anthracis* depends on two **virulence factors**: **polypeptide capsule** and **production of toxin (edema & lethal toxin)**
- Toxins produced consists of three proteins: Protective antigen (PA), lethal factor (LF) & edema factor (EF)
 - EF with PA forms **edema toxin**
 - LF plus PA form **lethal toxin**, which is a major virulence factor and cause of death in infected animals and humans

Clinical manifestations of Pulmonary anthrax....

- Pulmonary anthrax caused by inhalation of spore
 - Initial symptoms resemble common cold (malaise, mild fever, nonproductive cough).
 - Progress to severe breathing problems and shock
 - characterized by hemorrhagic pneumonia and lymphadenitis.
 - Usually results in death 1-2 days after onset of acute symptoms
 - Mortality rate 99% in untreated cases
 - Treatment usually not effective after symptoms are present

Laboratory diagnosis

- **Specimens** – pus or fluid or swabs from cutaneous lesions, sputum, blood and spinal fluid based on site of infections

Microscopic

- **Stained smears** from the local lesion or of blood from dead animals often show chains of large gram-positive rods with centrally located spores from culture
- Loeffler's polychrome (**McFadyean**) methylene blue stained smear: Square ended blue-black bacilli surrounded by a pink/purple capsule

Cultural tests

- **Culturing on blood agar:** the organisms produce **non-hemolytic gray to white mucoid colonies** with irregular borders
- **Biochemical test:** Ferment glucose, maltose, sucrose, trehalose, and dextrine, with acid production but not gas
- Positive for Nitrate reduction test, Catalase test, starch hydrolysis, VP, and gelatine liquefaction

Serological test

- ELISA measure antibodies against edema and lethal toxins

Prevention and control

□ *B. anthracis* is susceptible to penicillin.

□ Doxycycline and ciprofloxacin are alternative prophylactics

Prevention and control

- Disposal of animal carcasses by deep burial or burning
- Protective clothing and gloves for handling potentially infected materials Eg. Animal's hair, hide bone
- Active immunization of domestic animals with live attenuated vaccine
- Vaccination of workers in high risk occupation with anthrax toxoid

7. Mycobacterium tuberculosis

- Non- sporing, non-motile, strictly aerobic, Acid-fast bacilli
- The main reservoir is an infected human
- Have slow generation time (**18-24** hours)
- Once stained with primary stain (carbon fuchsine), they resist decolonization by 3% acid-alcohol
- Acid-fastness depends on the waxy envelope- mycolic acid of cell wall

Mycobacterium ...

- *M. tuberculosis* causes tuberculosis and is a very important pathogen of humans
- About 1/3rd of the world's population is infected with *M. tuberculosis*, with 30 million people having active disease
- Because of resistance to desiccations, the organisms can remain viable in the environment for a long time

Mycobacterium ...

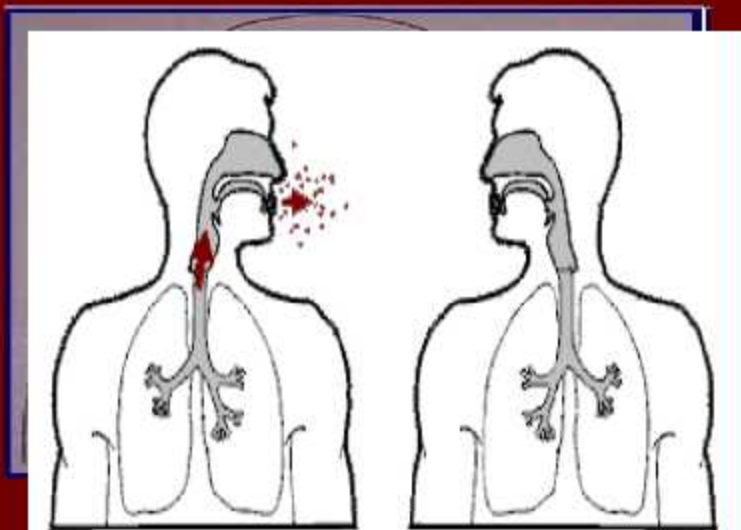
- ❑ M. tb does not possess the classic bacterial virulence factors such as **toxins, capsules** and **fimbriae**
- ❑ **Components of cell wall (Lipids):** Mycolic acid, waxes responsible for acid-fastness, granuloma formation and caseation necrosis
 - Protects the pathogen from antibiotic therapy and host defenses
 - Induce the immediate type of hypersensitivity
- ❑ **Proteins:** Elicit tuberculin reaction and antibody production

Transmission

- **Route of infection:** Respiratory- Inhalation of droplet nuclei and ingestion of infected milk

Respiratory spread

- Infection moves easily among people in close contact because it is spread primarily when infected droplets from the respiratory system circulate in the air due to coughing, spitting, or sneezing



Progression of active tuberculosis infection

□ Depending on the time of infection TB is classified into;

A. Primary tuberculosis – tubercle formation:

- Occurs in a person who has had no previous contact with the organism
 - The granulomatous lesions that develop known as tubercle
 - About 95%, the infection becomes arrested and 10% become progressive (active) infections
 - The only evidence of tuberculosis may be a positive **tuberculin test**
- Chest radiograph**

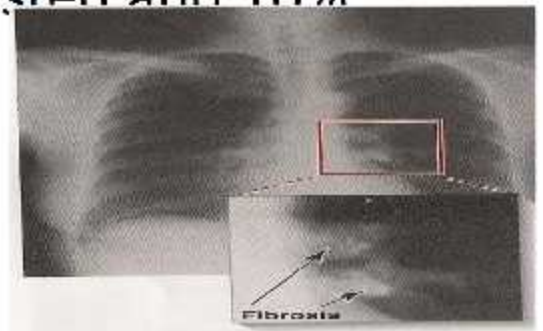


Figure 18.5
A chest radiograph showing some fibrosis – the classic Ghon complex.

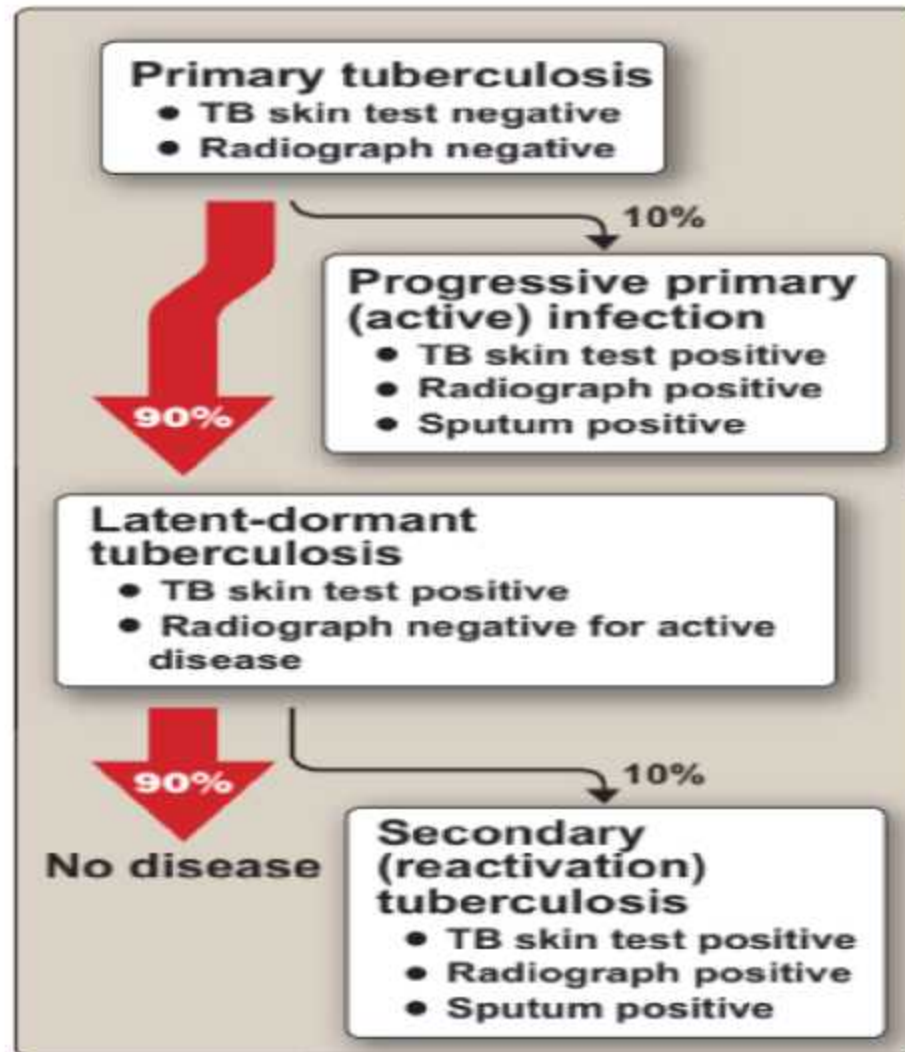
Progression of active tuberculosis infection.....

- If the lesion breaks down, the caseous (cheesy) material discharged and bacteria may spread to lymph, blood stream, lungs etc.
- Progressive disease may lead to;
 - Chronic pneumonitis
 - Tuberculous meningitis

B. Post primary tuberculosis

- Reactivation of latent infection which is usually seen in older individual and in those who develop impaired cell mediated immunity

Progression of active tuberculosis infection.....



Clinical significance

□ Tuberculosis has two major clinical forms: **Pulmonary and extra-pulmonary tuberculosis**

✚ **Pulmonary TB (80%)**

- Primarily occurs during childhood
- Secondarily 15-45 years or later
- Persistent cough for more than 3 weeks
- Productive cough with or without blood-stained sputum
- Shortness of breath and chest pain
- Intermittent fever, night sweat, loss of weight, loss of appetite, fatigue and malaise

Clinical significance.....

✚ **Extra pulmonary TB-** affects all parts of the body

—Most common sites are lymph nodes, pleura, gut, bone and joints, meninges and peritoneum

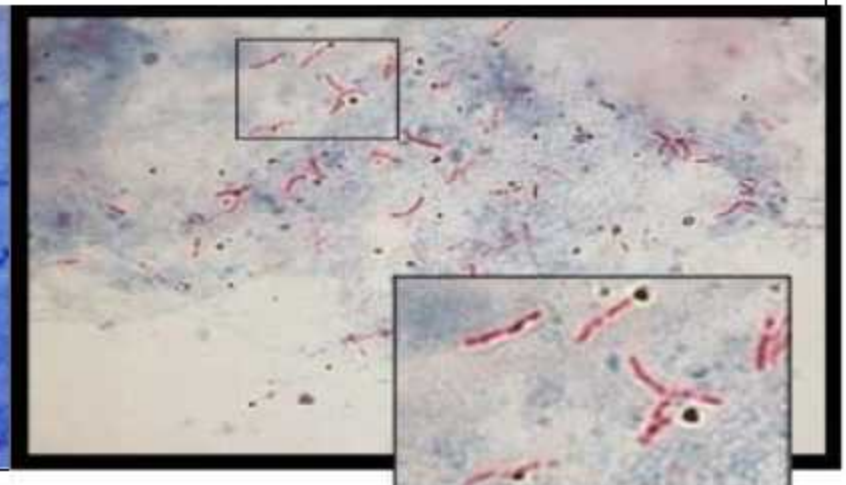
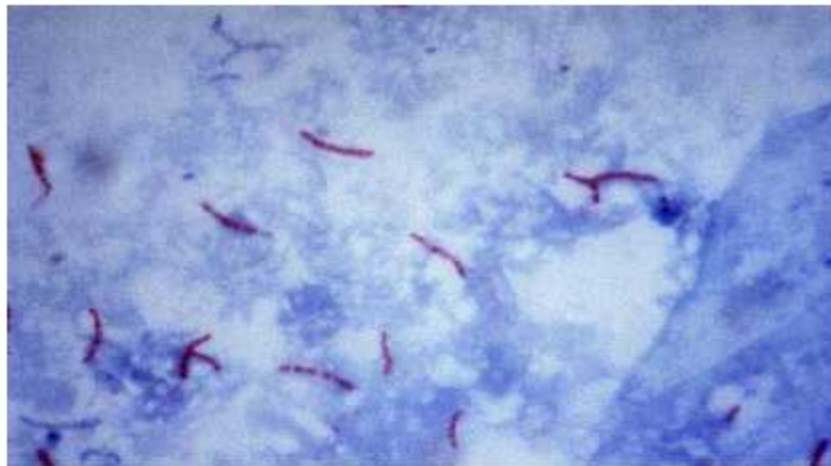
- TB lymphadenitis
- Tuberculosis pleurisy
- TB of bones and joints
- Tuberculous meningitis
- Intestinal TB
- Renal TB

Laboratory diagnosis

- Diagnosis of active pulmonary TB includes clinical symptoms and abnormal chest radiographs and confirmation by isolation of *M. tuberculosis* from relevant clinical material
- **Specimen:** Sputum, CSF, pleural and peritoneal fluid
- **Pulmonary tuberculosis:** Spot-Morning-Spot specimen of sputum is collected on two consecutive days (but now Spot-spot)

Laboratory diagnosis.....

1. **Ziehl – Neelsen staining or Acid fast staining:** Smear is made from the specimen on a new glass slide and stained by the Ziehl-Neelsen technique. It is examined under oil immersion lens.
 - Acid fast organisms – stained red
 - Back ground – blue



2. Auramine-phenol fluorochrome technique

- This technique can be used to detect *M. tuberculosis* in sputum, CSF etc.,
- Rapid (40X objective) and can detect few bacilli present in specimen when compared to Acid fast staining
- The fluorescent stained organisms can be seen glowing (fluorescing) against a **dark** background



Laboratory diagnosis.....

Culture: Lowenstein-Jensen (LJ) medium

- It is the ordinary selective media for tubercle bacilli
- Raised, dry, creamy colonies present after 3-8 wks of incubation at 37°C
- Acid fast bacilli is slow growing, non pigmented



Laboratory diagnosis.....

- **Serology:** Serology includes detection of anti-mycobacterial antibodies in patient serum with various methods such as ELISA, RIA and latex agglutination
- **Molecular methods:** It is a rapid method based on DNA amplification directly in clinical specimens
 - Routine PCR: Gene amplification
 - Real Time PCR: Gene amplification plus quantification
 - Sensitivity: 80 - 85%, specificity ~ 99%

Treatment

- Effective regimens must contain multiple drugs and usually given for 6-9 months
- Most commonly used **1st line of drugs** are;
 - Rifampin (RIF), or R
 - Isoniazid (INH), or H
 - Pyrazinamide (PZA) or Z
 - Ethambutol (EMB) or E
 - Streptomycin (SM) or S
- Adherence with this five-drug regimen is highly effective

Treatment.....

Second-line anti-tuberculosis drugs

- **If 1st line drugs failed:** MDR-TB (Isoniazid (INH) & Rifampicin)
 - Aminoglycosides (kanamycin, amikacin, capromycin)
 - Fluroquinolones (ciprofloxacin)
 - Thioamides (ethinoamide, prothionamide)
 - Cycloserine
 - Paraminosalicylic acid (PAS)
- There are no universally accepted standard regimens for the treatment of MDR-TB

Problems with TB treatment

□ Resistance to anti-TB drugs can occur When the drugs are misused

- Patients do not complete their full course of treatment
- Health-care providers prescribe the wrong treatment, the wrong dose, or length of time for taking the drugs
- Supply of drugs is not always available
- The drugs are of poor quality

Types of drug Resistance

- **Mono drug resistance:** resistance for one drug among the five group
- **Poly drug resistance:** resistance to more than one drug
- **Multidrug resistance (MDR):** is resistant to at least two of the best anti-TB drugs, isoniazid and rifampicin

Types of drug Resistance cont'd

- **Extensively Drug-Resistant (XDR):** is resistant to isoniazid and rifampin Plus resistant to any fluoroquinolones and at least one of three injectable second-line drugs i.e., Amikacin, Kanamycin, or Capreomycin

Prevention

- Early case detection & treatment
- Decreasing of over crowding
- Pasteurization of milk --- ↓ *M. bovis* infection
- Health education
- Immunization with BCG vaccine (**Bacillus of Calmette-Guerin**) which consists of a live attenuated strain of *M. bovis*
- It is not given for immunocopromized individual and it is not 100% effective. Studies suggest a 60-80% effective rate in children

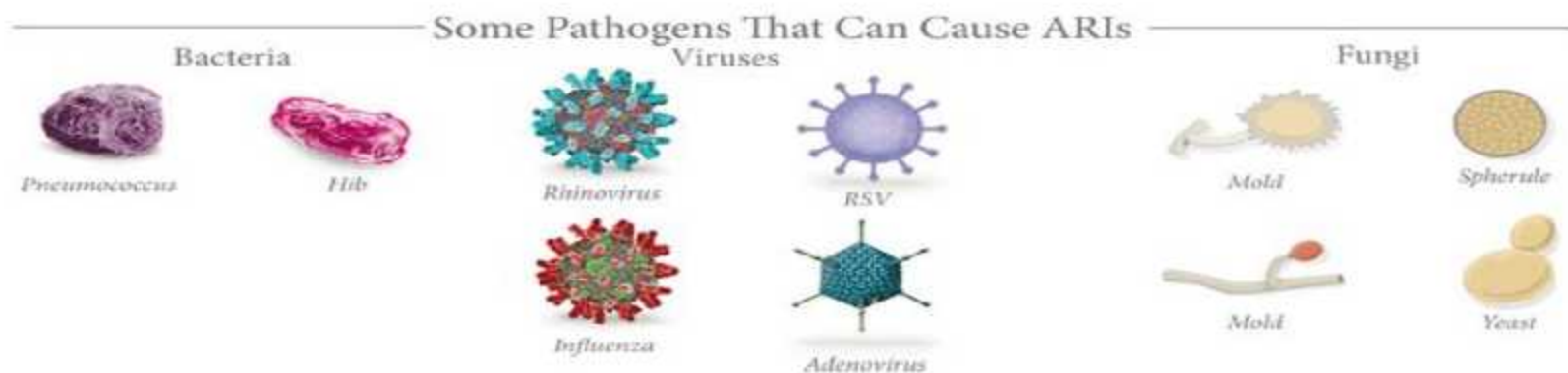
HIV Counseling

- All persons with suspected and confirmed TB disease should be offered HIV counseling and blood testing, in addition to treatment
- This is because TB is more likely to occur among HIV positive individuals



IV. Viral Disease of the Respiratory System

- Influenza virus
- Para-influenza virus
- Respiratory Syncytial Virus (RSV)
- Adenovirus
- Rhinoviruses
- Coronavirus
- Human Metapneumovirus



Viral Pneumonia.....

- A number of viruses can cause pneumonia as a complication of infections such as with viral influenza.
- The etiologies of viral pneumonia are not usually identified in a clinical laboratory because of the difficulty in isolating and identifying viruses.
- Therefore, if no other cause is found, for example negative bacteria and fungus cultures and *Mycoplasma pneumonia* has been ruled out, a viral etiology may be assumed.

Viral Pneumonia.....

- The most common sources of **viral pneumonia** are **adenoviruses, influenza viruses, Parainfluenza viruses,** and **respiratory syncytial viruses.**
- The signs and symptoms produced by these viruses can range from mild cold-like symptoms to severe cases of pneumonia

1. Influenza virus

- Influenza viruses cause respiratory infections worldwide; there are three types: A, B and C.
- Influenza A is found in aquatic birds, poultry and pigs, and these play important roles in human infections
- Influenza viruses are an enveloped, segmented ssRNA viruses that **regularly undergo genetic change**.
- The envelope contains two glycoprotein spikes: **Neuraminidase (NA)** and **haemagglutinin (HA)**, can show gradual **genetic** change (antigenic drift) and sudden major change (**antigenic shift**).

Antigenic drift and shift

- NA & HA are important antigens that determine antigenic variations in influenza viruses.
- Both glycoprotein frequently undergo genetic changes decreasing the effectiveness of the host immune response.
 - **Antigenic drift:** Minor antigenic changes that occur due to mutation.
 - **Antigenic shift:** Major antigenic changes (when HA or NA or both change) which results in new strains of the virus.
 - **It has been identified only in influenza virus type A**

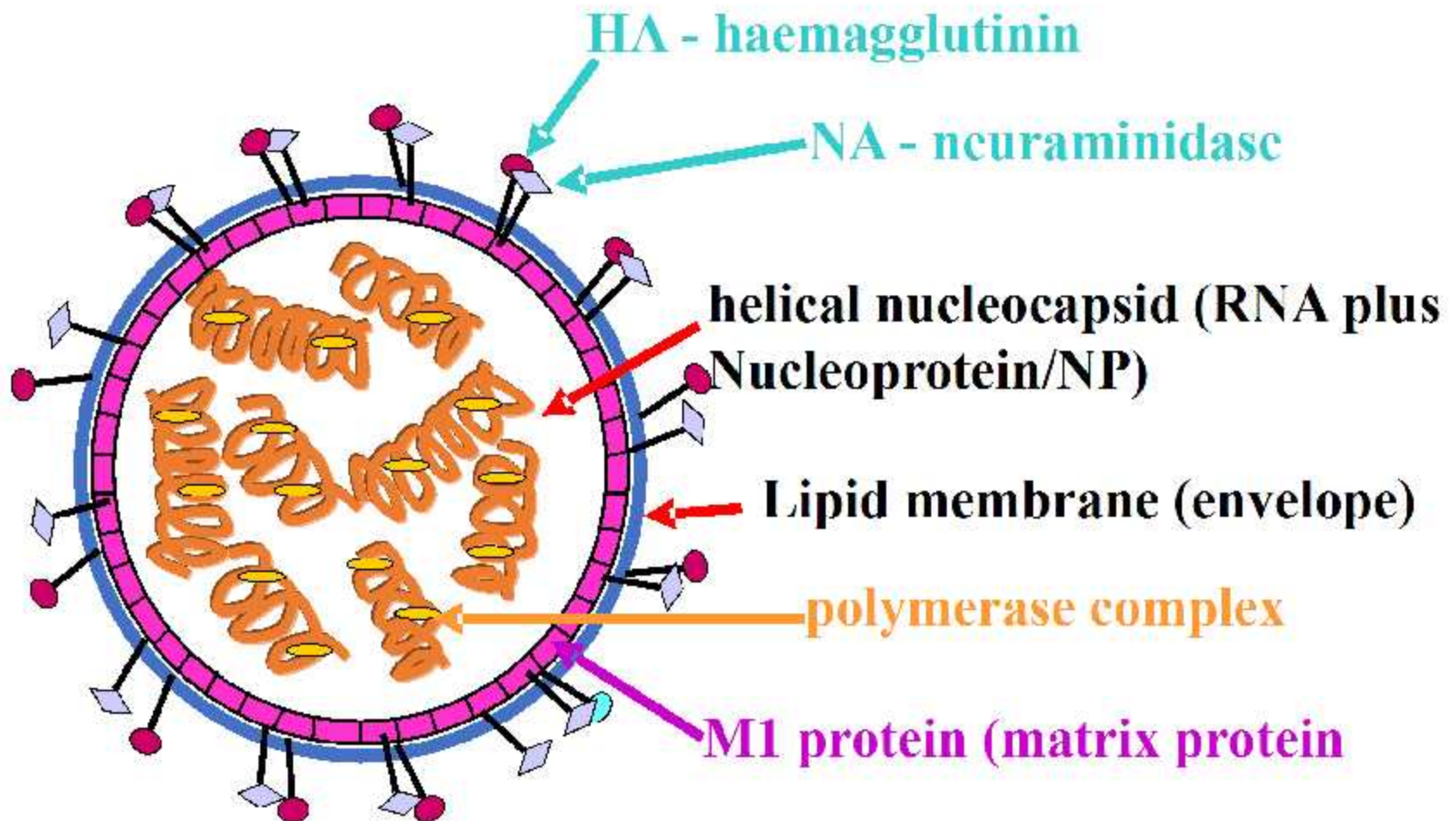


Figure: Influenza virus structure

Pathogenesis and disease

- The influenza virus is primarily transmitted by direct contact and inhalation of aerosols.
- The virus attaches and multiplies locally in the mucous membrane of the respiratory tract causing **rhinitis, pharyngitis & bronchitis**.
- Manifestations are fever, chills, myalgia (muscle aches), headache, dry cough and anorexia.
- The disease is self-limiting, however, it may be complicated by bacterial pneumonia due to *H. influenza*, *S. aureus* and *S. pneumonia*.

Pathogenesis and disease.....

- A complication of influenza that occurs primarily in children and teenagers is **Reye syndrome**.
- This sequela causes swelling in the liver and brain, and may progress to neurological damage, coma, or death.
- Influenza B & C is known to cause only minor respiratory disease; probably not involved in epidemics.

Laboratory Diagnosis

- Viral isolates are identified by hemagglutination-inhibition tests and immunofluorescence testing with monoclonal antibodies.
- Isolation - tissue culture.
- Detection of genome - molecular techniques
- Serology - ELISA, RIA, hemagglutination inhibition test.

2. Parainfluenza virus

- Respiratory syncytial virus, parainfluenza viruses and human metapneumovirus are important and frequent causes of respiratory tract infections, especially in children.
- Parainfluenza virus is a major cause of respiratory disease in infants and young adults and mostly associated with: **croup**.
- The virus belongs to the paramyxovirus group and it is single-stranded enveloped RNA viruses
- Infection is transmitted with droplets.

Parainfluenza virus.....

- Transmission and pathology similar to influenza virus, but there are differences.
 - Parainfluenza virus replicates in the cytoplasm.
 - Influenza virus replicates in the nucleus.
- **Respiratory disease: 'common cold'** syndrome due to infection of the nose and throat, or it may cause severe laryngotracheobronchitis

3. The Common Cold viruses

- More than 200 different viruses are known to cause the common cold.
- The most common groups of cold viruses include **rhinoviruses (50%), coronaviruses (15-20%), and adenoviruses.**
- Extremely small, non-enveloped, single-stranded RNA viruses
- Symptoms include sneezing, nasal secretions, congestion, sore throat, coughing.

Common Cold viruses.....

- Sinus infections, lower respiratory tract infections, laryngitis, and otitis media can occur as complications of a cold.
- Colds are most often transmitted through direct contact and droplet transmission
- Coughing and sneezing produce infectious aerosols, and rhinoviruses are persist on environmental for up to a week
- The absence of high fever is typically used to differentiate common colds from other viral infections, like influenza.

4. Adenoviruses

- Non – enveloped, Icosahedral capsid, ds DNA virus
- More than 100 serotypes of which at least 47 infect humans
- Clinical syndrome is different based on organ or system involved
 - **Upper respiratory infections:** common cold (rhinitis); pharyngitis & tonsillitis
 - **Lower respiratory infections:** bronchitis; acute respiratory disease & pneumonia

Adenoviruses.....

□ In addition, they cause infections of;

□GIT

□Eye

□Genito- urinary tract

5. Respiratory Syncytial Virus (RSV)

- RSV is the most common **cause of pneumonia in infants** with many hospitalizations and deaths/year.
- Epidemics appear in winter/early spring.
- Cough and sneezing for > 1 week and fever if bacteria become involved.
- Rapid testing using serology and Nearly 100% show evidence of infection by age 2 years
- **Ribavirin** may be given in aerosol to diminish symptoms.

SARS

- Severe acute respiratory syndrome (**SARS**) is acute respiratory infections caused by **coronaviruses**.
- these are thought to be zoonotic infections.
- Bats and civet cats are thought to be the reservoirs for SARS
- Signs and symptoms include high fever, headache, body aches, and cough, and most patients will develop pneumonia.

- COVID-19: Spherical or pleomorphic enveloped particles containing single-stranded (positive-sense) RNA associated with a nucleoprotein within a capsid comprised of matrix protein. The envelope bears club-shaped glycoprotein projections.

Viral Respiratory Diseases Causing Skin Rashes

- Measles, rubella (German measles), and chickenpox are three important viral diseases often associated with skin rashes.
- However, their symptoms are systemic, and because their portal of entry is the respiratory tract, they can be considered respiratory infections.

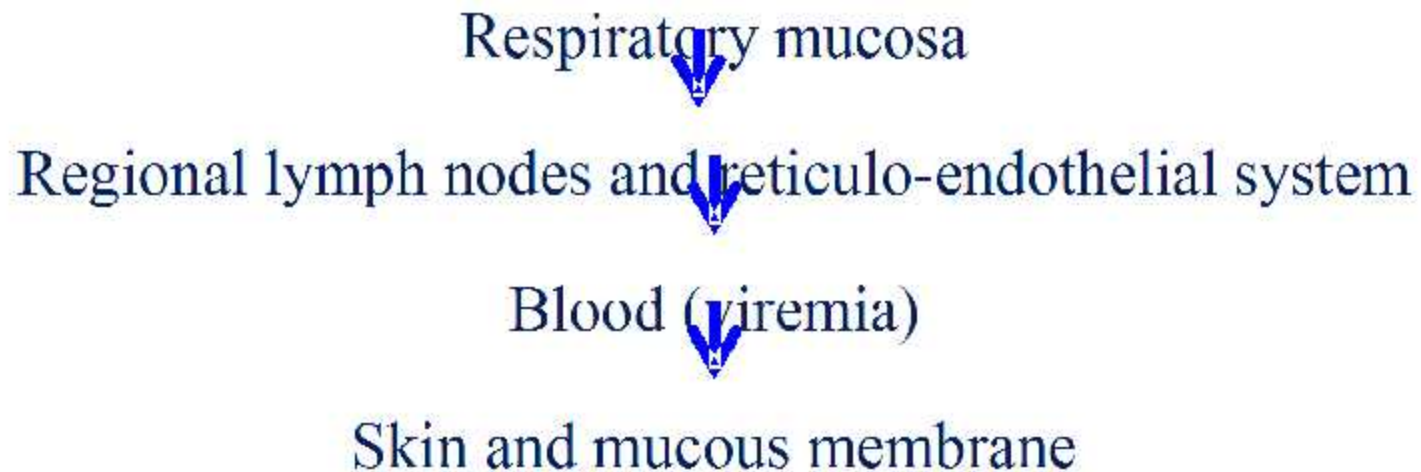
Mumps virus

- Infection is probably acquired by inhalation of droplets into the respiratory tract.
- The incubation period is 14–18 days, and is followed by a generalized illness with later localization in the salivary glands, usually the parotids.
- The generalized phase is the usual **'flu-like'** illness with fever, followed by developing pain in the **parotid glands, which then swell rapidly.**

Morbillivirus (measles virus)

- The virus causes **measles**, also known as **red measles** and **rubeola**, a highly infectious disease that affects children.
- It is characterized by **fever** and **maculopapular skin rash**.
- Very contagious; transmitted by respiratory aerosols.
- Humans are the only reservoirs.
- Frequent cause of death worldwide.
- Virus invades respiratory tract.

Pathogenesis



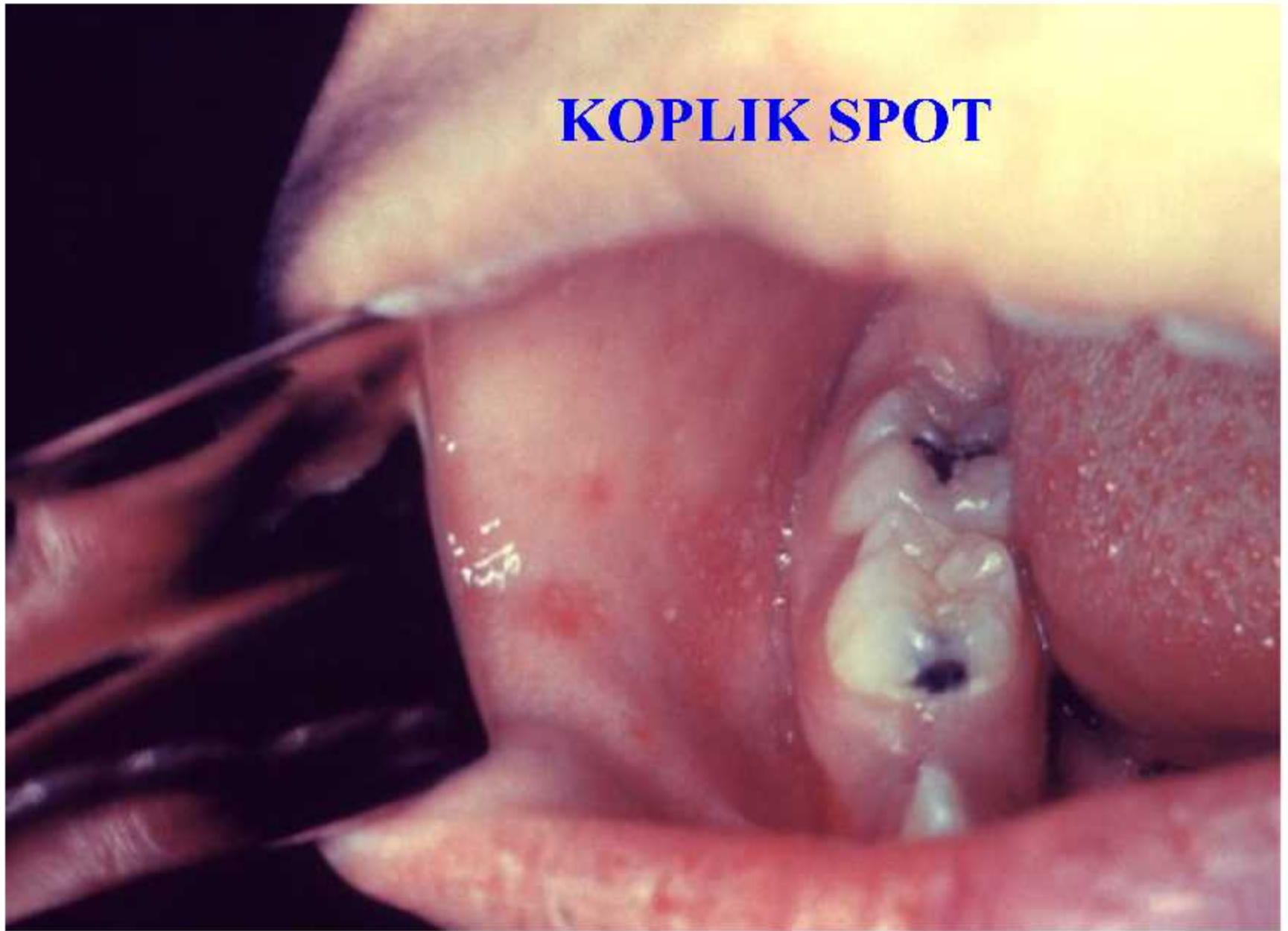
- Most serious complication is subacute sclerosing panencephalitis (SSPE), a progressive neurological degeneration of the cerebral cortex, white matter and brain stem.

Clinical Manifestations

- Measles is initially characterized by a high fever, conjunctivitis, and a sore throat, dry cough, lymphadenitis, coryza (running nose), photophobia.
- The virus then moves systemically through the bloodstream and causes a characteristic maculopapular skin rash.
- Another diagnostic sign of measles infections is **Koplik's spots**-oral lesions (red spots with bluish-white specks in their centers appearing in the oral mucosa opposite the second molars).
- Long lasting immunity after a single infection.
- Attenuated viral vaccine **MMR** is available for future protection.



KOPLIK SPOT



VI. Fungal Diseases of the Respiratory System

- Fungi are also a source of respiratory infection.
 - Usually in immunocompromised patients
 - Most dangerous are *Aspergillus* and *Pneumocystis*.
- Fungal spores are easily inhaled; they may germinate in the lower respiratory tract.
- The mycoses below can be treated with imidazole drugs (Itraconazole) and amphotericin B (serious life threatening cases as this drug is very toxic).

1. Coccidioidomycosis

- ▣ Coccidioidomycosis is respiratory infection caused by ***Coccidioides immitis***.
- ▣ Infection is acquired through inhalation of air born arthroconidia.
- ▣ Infection is usually self limiting.
- ▣ Disease is manifested as an influenza-like illness with fever, malaise, cough, arthralgia and headache.

Coccidioidomycosis.....

- Infection can be symptomatic or asymptomatic.
 - Symptomatic form known as Valley Fever.
- Disseminated coccidioidomycosis is seen in patients with AIDS and on immunosuppressive therapy.
- Can also cause a form of coccidioidal meningitis
 - Can be fatal if not treated aggressively

Treatment

- Most infections are self limiting and supportive treatment is sufficient.
- **For immunocompromised individuals:** oral ketoconazole, oral fluconazole or IV amphotericin B followed by oral antifungal drugs.

2. Histoplasmosis

- ▣ Dimorphic fungus: grows as mycelium in culture, as small budding yeasts in host
- ▣ Histoplasmosis is an intracellular mycosis of the RES caused by ***Histoplasma capsulatum***-Dimorphic fungus grows as mycelium in culture, as small budding yeasts in host
- ▣ Occurs in soil contaminated with bat or bird droppings
- ▣ Infection is acquired through the respiratory tract by inhalation of conidia.
 - ▣ Commonly found in temperate, subtropical, and tropical zones
 - ▣ 50% - 90% of residents in these areas test positive for exposure.

Histoplasmosis ...

- ▣ Self limiting flu-like syndrome (95%) with fever, chills, myalgia, headache and non-productive cough are observed in symptomatic cases.
- ▣ chronic pulmonary disease mimics TB; most common in settings of prior lung damage
- ▣ Severe, dissemination occur in immunosuppressed individuals.
 - ▣ Lymphadenopathy, enlarged spleen and liver, anemia and high mortality are observed in the absence of treatment.

Treatment

- For primary pulmonary Histoplasmosis: supportive treatment and rest
- For mild to moderate infection: oral ketoconazole
- In disseminated disease: amphotericin B
- In patients with AIDS: Amphotericin B followed by oral ketoconazole

3. Blastomycosis

- ▣ Blastomycosis is caused by *Blastomyces dermatitidis*.
- ▣ Infection is acquired via inhalation of the conidia.
- ▣ Pulmonary infiltration with symptoms like fever, malaise, night sweats, cough and myalgia are observed.
- ▣ When dissemination occurs, skin lesions on the exposed surface are common.
- ▣ There is **no** increased incidence in immunosuppressed patients.

Blastomycosis: Pathogenesis

- Infection of the lungs is gradual.
 - Fever, chills, and drenching sweats develop.
 - Chest pain, difficulty breathing, and cough may also develop.
 - Can sometimes heal without treatment.
 - Skin – warty patches develop surrounded by tiny painless abscesses
 - Bones – painful swellings
 - Genitounrinary tract – prostatitis or painful swelling of epididymis

Treatment

- Oral ketoconazole or itraconazole for 6 months
- Amphotericin B intravenously

4. Paracoccidioidomycosis

- ▣ The causative agent is *Paracoccidioides brasiliensis*.
- ▣ Infection is acquired via inhalation and early lesions occur in the lung.
- ▣ Dissemination occurs primarily in the spleen, liver, mucus membrane and skin.
- ▣ Enlarged lymph node and gastrointestinal symptoms may be the only presenting symptoms.

Treatment

- Mild to moderate disease: oral sulfonamides or ketoconazole
- Disseminated disease: Amphotericin B

5. Aspergillosis

- Caused by the fungus *Aspergillus*
 - Widely distributed and found throughout world
 - Infection is through inhalation of resistant conidia.
 - Seen more and more in nosocomial infections associated with air-conditioning systems.
- Invasive Aspergillosis shows a rapid progression to death.

Aspergillosis.....

- Typically seen in the immunocompromised.
 - Particularly patients with leukemia or AIDS.
 - Patients undergoing bone marrow transplantation.
- Also seen in individuals with preexisting pulmonary disease
 - Chronic bronchitis, asthma, and tuberculosis
- Fungus produces extracellular proteases, phospholipases, and toxic metabolites.

..Aspergillosis: Pathogenesis

- Colonization with *Aspergillus* leads to invasion of lung tissues w/c causes penetration of blood vessels.
 - This causes hemoptysis and/or acute pneumonia.
- Pneumonia is accompanied by multifocal pulmonary infiltrates and high fever.
 - Mortality for invasive Aspergillosis is 100%.
 - Amphotericin B and itraconazole can be used but are usually ineffective.

6. *Pneumocystis carinii* Pneumonia

- ▣ *P. carinii* (*jirovecii*) was thought to be a protozoa until recently.
- ▣ *Pneumocystis jiroveci*, currently classified as a fungus, is found in healthy human lungs
- ▣ It occurs in two distinct morphologic forms: trophozoites and cysts.
- ▣ The organism is an extracellular pathogen.

Pneumocystis carinii Pneumonia ...

- ▣ In the lung, growth is limited to the surfactant layer above alveolar epithelium.
- ▣ Causes disease in immunosuppressed patients.
Transmitted by direct contact.
- ▣ Causes alveoli to become filled with exudate and leads to **interstitial plasma cell pneumonitis**.
- ▣ Untreated infections maybe lethal in these cases.

Laboratory diagnosis & Treatment

- **Sample:** bronchoalveolar lavage, lung biopsy
 - Giemsa, silver or toluidine blue staining

Treatment

- Trimethoprim sulfamethaxazole (oral or sometimes intravenous) is given for acute treatment.

1 Spore production

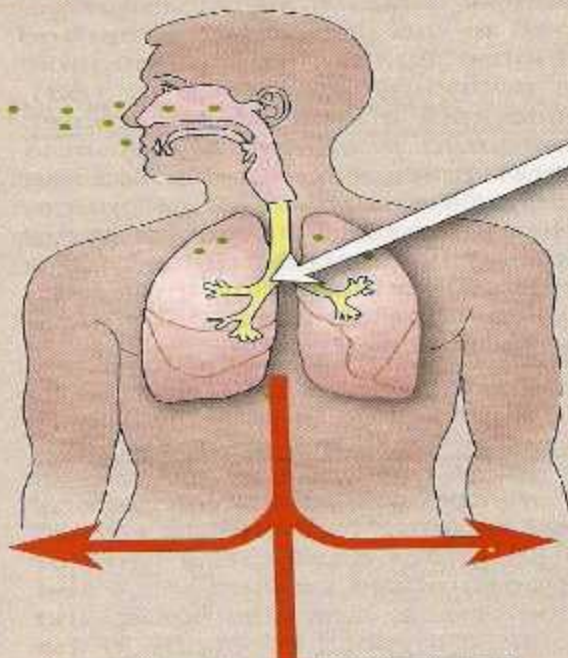
Fungi are found in soil and the feces of birds and bats. The fungi produce spores that become airborne.



Inhalation
of spores

2 Primary pulmonary infection

When inhaled, spores cause a primary pulmonary infection.



Lung of a patient with acute coccidioidal pneumonia

Coccidioidomycosis



Spherule

Possible sites of infection:

- Central nervous system
- Bone

Methenamine silver stain

Histoplasmosis



Possible sites of infection:

- Liver
- Spleen
- Lymph nodes
- Bone marrow

Yeast form of *Blastomyces dermatitidis*

Blastomycosis

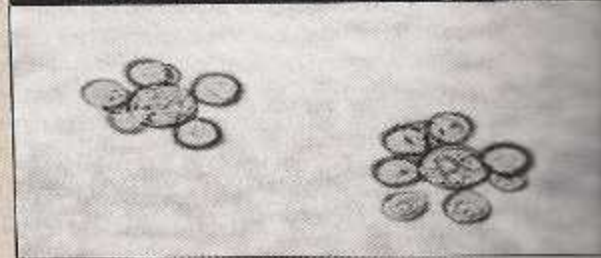


Possible sites of infection:

- Skin
- Bone
- Genitourinary tract

Central fungal cell with a series of buds that look like spokes of a wheel

Paracoccidioidomycosis



Possible sites of infection:

- Mucosa of the mouth and nose

3 Dissemination of infection

The fungi may travel from the lungs to other sites where infection can occur.



End of this Chapter!!!